

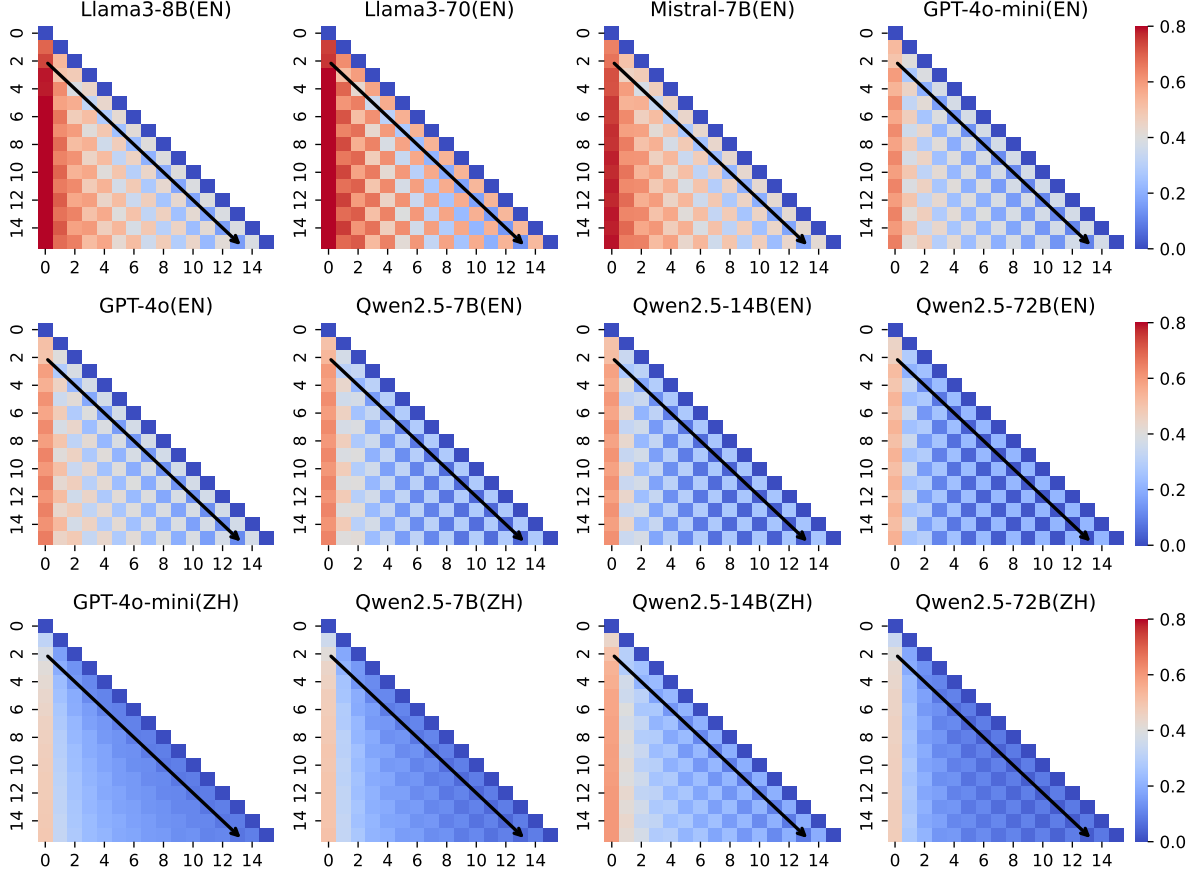


Figure 2: The difference confusion matrix for successive paraphrasing, where EN and ZH denotes English and Chinese sentence-level paraphrase generation accordingly. Both the x and y axes represent paraphrases at each step, and the value at the $(i\text{-th}, j\text{-th})$ grid position indicates the difference between the paraphrases at the $i\text{-th}$ and $j\text{-th}$ positions. A darker color indicates a smaller difference value between two paraphrases. The black arrow underlines the differences between T_i and T_{i-2} , and averaging these values and subtracting the result from 1 gives our 2-period degree τ .

between two paraphrases. For instance, if successive paraphrases exhibit perfect 2-periodicity such that $d(T_i, T_{i-2}) = 0$, then $\tau = 1$, indicating that the current paraphrase matches exactly with that from two steps earlier. To evaluate semantic equivalence, we employ cosine similarity on sentence embeddings¹ (Reimers and Gurevych, 2019).

4 Results

Building on the dynamical systems perspective introduced earlier, we now examine the empirical evidence that successive paraphrasing leads LLMs toward stable attractor cycles. We iteratively paraphrase sentences over 15 rounds within the sentence-level dataset and calculate the 2-periodicity degree.

¹<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

4.1 Periodicity

We calculate the textual difference between T_i and T_{i-2} for paraphrases at each step. Arranging these differences into a confusion matrix (Figure 2) reveals a pronounced 2-period cycle. For all LLMs, the matrix’s alternating light and dark patterns indicate that paraphrases generated at even iterations cluster together, and similarly, those at odd iterations form another cluster. This clear partitioning aligns with the behavior of a dynamical system converging onto a 2-period limit cycle—an attractor that draws the iterative process into a stable oscillation between two distinct states.

We also quantify this periodicity across different LLMs, as shown in Table 1. While all models exhibit some degree of 2-periodicity, Qwen2.5-72B shows a particularly strong and consistent cycle in both English and Chinese, whereas Llama3-70B displays relatively weaker periodic behavior. Mod-

Mistral-7B	Llama3-8B	Llama3-70B	GPT-4o-mini	GPT-4o	Qwen2.5-7B
0.71	0.72	0.60	0.83	0.81	0.86
Qwen2.5-14B	Qwen2.5-72B	Qwen2.5-7B	Qwen2.5-14B	Qwen2.5-72B	GPT-4o-mini
0.89	0.92	0.70	0.84	0.92	0.88

Table 1: The periodicity degree τ of different LLMs. The models represented in blue denotes the English paraphrase generation, while red indicating Chinese paraphrasing.

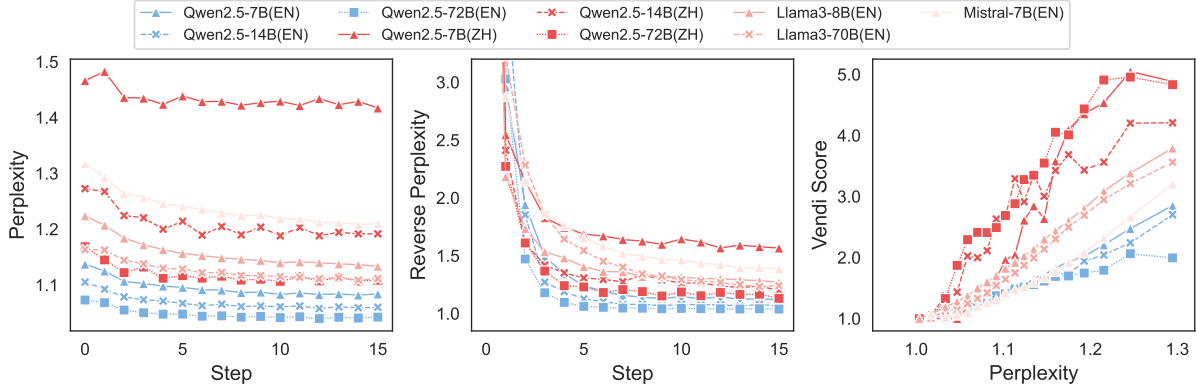


Figure 3: Convergence of perplexity, reverse perplexity, and generation diversity. The left and middle plots show that as the number of steps increases, both perplexity and reverse perplexity decrease steadily until they reach their lower bounds. The right plot shows that generation decreases as perplexity decreases.

els with higher periodicity tend to retain more semantic fidelity, suggesting that the recurring attractor states preserve core meaning even as they oscillate between two paraphrastic forms, as shown in Appendix B.

While this periodicity can be viewed as an implicit repetition issue, it differs from explicit repetition of previously seen context. Instead, the model implicitly cycles through a limited set of paraphrastic forms without directly referencing prior iterations. In terms of systems theory, the model’s mapping function P creates a dynamical environment in which the state space is not fully explored, with the trajectories settling into a 2-period attractor.

4.2 Convergence to Stable Attractor

To probe the internal dynamics that lead to these attractor cycles, we explore generation determinism with successive paraphrasing unfolds. We define a **conditioned perplexity** $\sigma(T_i | T_{i-1})$, reflecting the model’s confidence in generating T_i given T_{i-1} , and a **reverse perplexity** $\hat{\sigma}(T_i | T_{i+1})$, indicating how easily T_i could be reconstructed from T_{i+1} .

Figure 3 demonstrates that as successive paraphrasing proceeds, both perplexity and reverse perplexity decrease. The forward direction (perplexity) quickly converges to a low boundary, while the

reverse direction starts high, indicating that initially it is hard to “go back” from T_{i+1} to T_i . However, it drops fast as paraphrasing proceeds and aligns with the forward perplexity. Finally, the system evolves towards a state where generating T_{i+1} from T_i is nearly as deterministic and predictable as reconstructing T_i from T_{i+1} . This symmetry resembles a **stable attractor** in a dynamical system, where bidirectional predictability indicates that the system has “locked in” to a limit cycle.

We further quantify generation diversity by sampling multiple paraphrases at each iteration and computing the Vendi score (Friedman and Dieng, 2022). As shown in Figure 3, a low perplexity indicates a low generation diversity. A Vendi score of one indicates that all paraphrases in the beam are identical to each other. As both forward and reverse perplexity decreases, the model consistently produces similar paraphrases, leaving minimal room for alternative textual trajectories. From a systems viewpoint, the collapse into low perplexity and low diversity states corresponds to the model settling into the basin of attraction of a periodic orbit. Once inside the basin, the model’s generative behavior becomes nearly deterministic, causing the output sequence to cycle predictably.

The notion of invertibility, where each para-

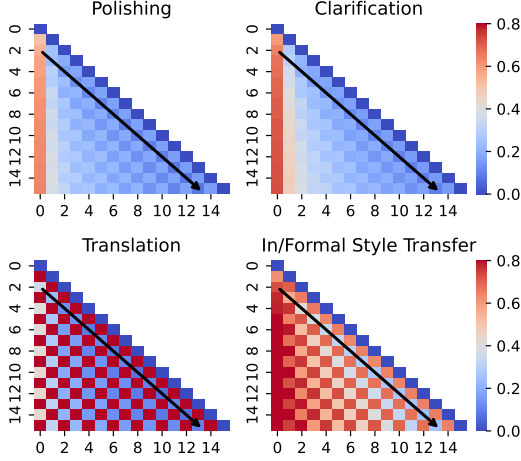


Figure 4: The difference confusion matrix for four tasks beyond paraphrasing. Note that in translations, the difference between texts in two different languages is set to one.

phrase can be treated as a paraphrase of its own paraphrase, further explains the robustness of periodicity. Invertibility places constraints on the mapping function P , effectively enabling a bidirectional relationship between states which encourages stable cycles. This insight suggests that tasks with similar invertible properties, e.g., translation, can also display limit cycle behavior, a hypothesis we will explore in Section 5.1.

5 Analysis

In this section, we perform analytical experiments on paragraph-level paraphrase datasets to generalize our findings to longer texts. We first demonstrate the extension of our findings to other task formats (Section 5.1). Then we go through a set of methods to try to escape from the attractor cycles in the remaining subsections.

5.1 Beyond Paraphrase Generation

Our earlier results indicate that successive paraphrasing leads LLMs to settle into periodic attractors—specifically, 2-period limit cycles. According to the systems-theoretic perspective, such cycles should arise whenever the transformation is invertible, enabling a bidirectional mapping that makes prior states easily reproducible. To test this, we examine four additional invertible tasks at the paragraph level: polishing (Pol.), clarification (Clar.), informal-to-formal style transfer (I/F.), and forward/backward translation (Trans.). These tasks are defined in Appendix C.1.

Figure 4 shows that even for these varied tasks, LLMs repeatedly converge to stable states, exhibiting pronounced 2-periodicity. Table 2 shows the degree of 2-periodicity across these tasks, with values ranging from 0.65 to 0.87. This finding reinforces the idea that invertibility fosters the emergence of limit cycles, as the model iterates the transformation and settles into an attractor. While paraphrasing is our primary lens, these findings confirm that stable attractor cycles are a broader characteristic of LLM behavior in iterative, invertible mappings.

Tasks	Para.	Clar.	Pol.	I/F.	Trans.
τ	0.80	0.83	0.86	0.65	0.87

Table 2: Impact of perturbations on periodicity compared to the original during paraphrasing.

5.2 Alternating Models and Prompts

One intuitive approach to escape an attractor is to introduce perturbations in the transformation itself. We attempt this by varying both models and prompts during successive paraphrasing. For **prompt variation**, we design four different paraphrasing prompts (refer to Appendix C.2) and randomly select one at each iteration. Despite regularly switching prompts, the 2-period cycle persists, as shown in Figure 5.

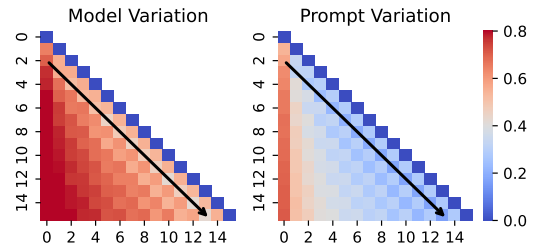


Figure 5: The difference confusion matrices for model variation and prompt variation.

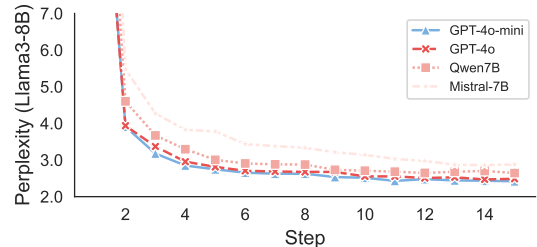


Figure 6: The perplexity of T_i conditioned on T_{i-1} calculated by Llama3-8B. Both T_i and T_{i-1} are generated by other LLMs.

Similarly, we introduce **model variation** by al-